Wiring Diagram — Metal Spinning Machine

PLC: Siemens S7-1214C (TIA Portal V17) **Scope:** Full system — power distribution, safety circuit, motion drives/VFD, PLC digital/analog I/O, pneumatic solenoid manifold.

This document is the field-wiring reference for the machine. Diagrams are written as **Mermaid** (renders inline on GitHub and in VS Code with a Mermaid extension); Sheet 3 also includes a **Graphviz DOT** block for users who prefer DOT.

How to read this document

- **Address authority:** All PLC addresses come from the TIA Portal tag export `Program/docs/PLCTags.xlsx` and are cross-checked against the I/O references in `Program/08_Main_OB1.scl`. The SCL code uses symbolic tag names only — the addresses live in the PLC tag table.
- **Generic blocks:** Power-side hardware (24 V PSU, safety relay, VFD, stepper/servo drives, motors, contactors) is **not** defined in the PLC code. It is drawn as **generic blocks**. ⚠️ **Confirm every generic block against the actually installed panel hardware before wiring.**
- **Contact logic:** Pay attention to NC vs. NO/PNP logic and fail-safe states — see the Legend.

⚠️ Signals referenced in code but missing from the tag export (assign during commissioning)

These inputs are used in `08_Main_OB1.scl` but were **not present** in the `PLCTags.xlsx` export. Assign real %I addresses during commissioning and update this document:

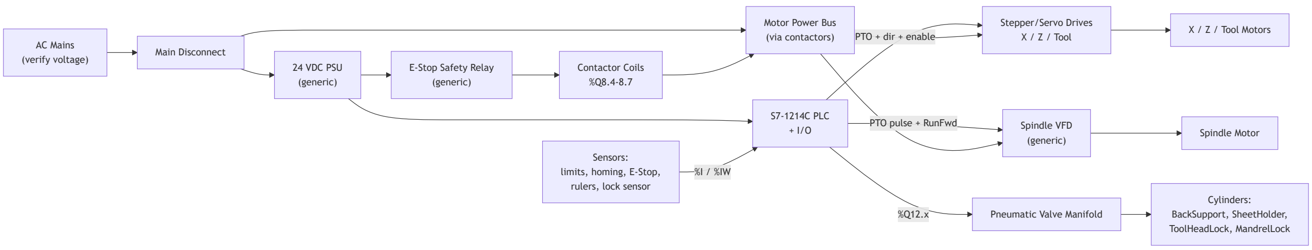
Signal	Used in	Suggested type	Address
`Panel_Stop`	OB1 fbProcess.Panel_Stop	DI (button / HMI)	**TBD**
`Panel_Pause`	OB1 fbProcess.Panel_Pause	DI (button / HMI)	**TBD**
`Panel_Reset`	OB1 fbProcess.Panel_Reset	DI (button / HMI)	**TBD**
`Safety_Door`	OB1 fbProcess.Door_Closed	DI (NC door switch)	**TBD**
`Safety_Air`	OB1 fbProcess.AirPressure_OK	DI (pressure switch)	**TBD**

> Stop / Pause / Reset are operated from the HMI in the current build (see `HMI_Tag_Guide.md`, `DB_HMI.Btn_*`). If physical panel buttons are also wired, give them the addresses above.

Legend / Símbolos

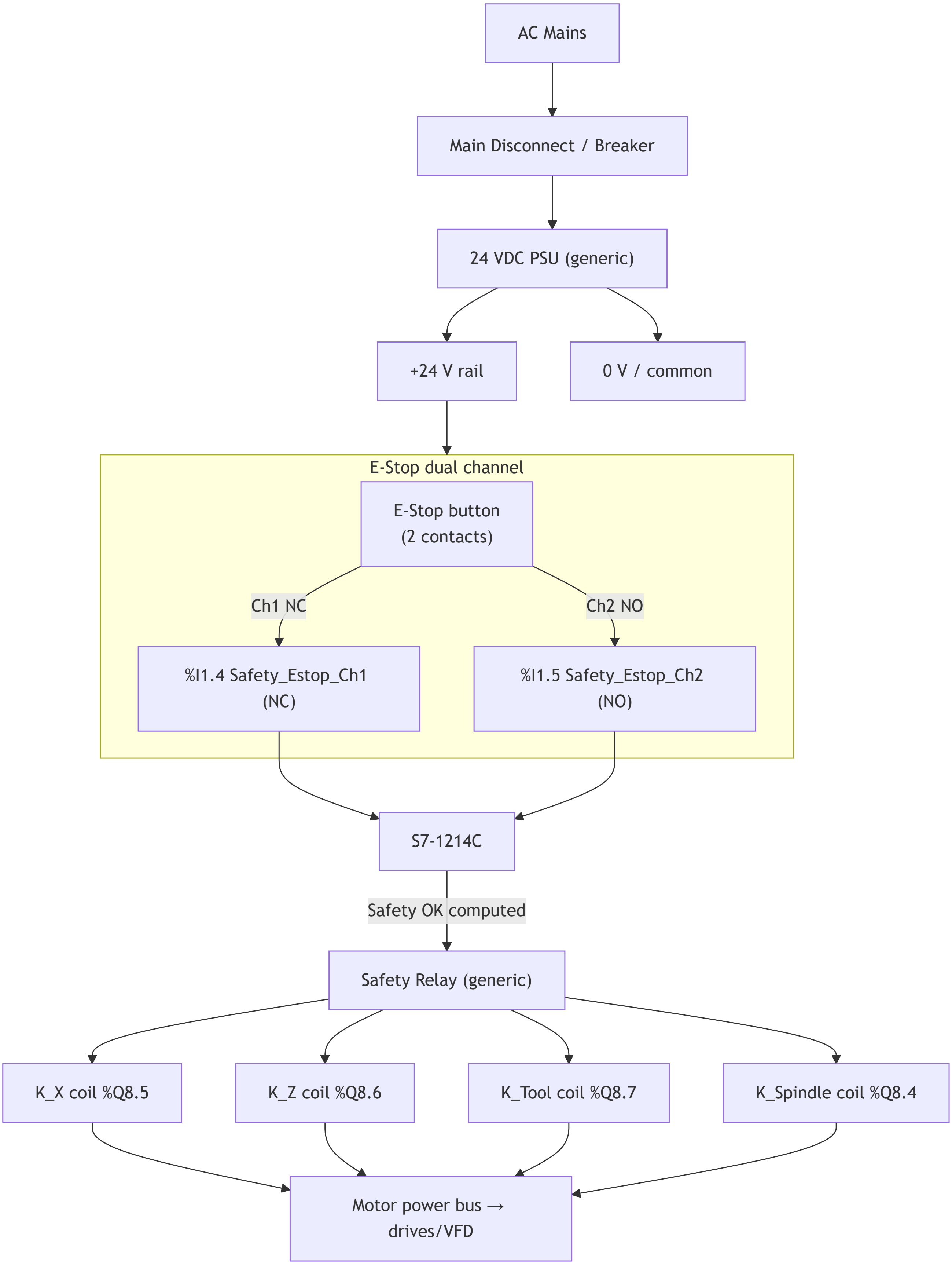
Notation	Meaning
`%I x.y`	PLC digital input bit
`%Q x.y`	PLC digital output bit
`%IW n`	PLC analog input word
NC	Normally-closed contact — TRUE/closed = safe; opens on fault (fail-safe)
NO / PNP	Normally-open / PNP proximity — TRUE = target/zone detected
PTO	Pulse-train output (high-speed) — pulse + direction pair per axis
5/2 SR	5-port 2-position, single solenoid, spring return (de-energize = spring side, safe)
5/3 BC	5-port 3-position, blocked center (holds position when both solenoids off)

Sheet 0 — System Overview / Diagrama de Bloques



Sheet 1 — Power Distribution & Safety Circuit

⚠️ PSU, safety relay, and contactors are generic — confirm models, coil voltage, and contact ratings against the installed panel.



Notes

- The PLC evaluates the dual E-Stop in `FB_EStopDualChannel` (runs first in OB1). Channels must be **opposite** (Ch1 NC closed AND Ch2 NO open) = safe. A mismatch lasting > 500 ms raises **0x0406 — E-Stop contact discrepancy** (check wiring/contacts).

- Contactor coils are driven by `FC_ContactorControl` outputs only when E-Stop is OK and the machine is not in STOPPED. Wire the safety relay so loss of the safety circuit also drops motor power independently of the PLC output (hardware fail-safe).

Signal	Address	Device
Safety_Estop_Ch1	%I1.4	E-Stop button, channel 1 (NC)
Safety_Estop_Ch2	%I1.5	E-Stop button, channel 2 (NO)
Output_Contactor_Spindle	%Q8.4	Spindle VFD power contactor coil
Output_Contactor_X	%Q8.5	X motor power contactor coil
Output_Contactor_Z	%Q8.6	Z motor power contactor coil
Output_Contactor_Tool	%Q8.7	Tool turret motor power contactor coil

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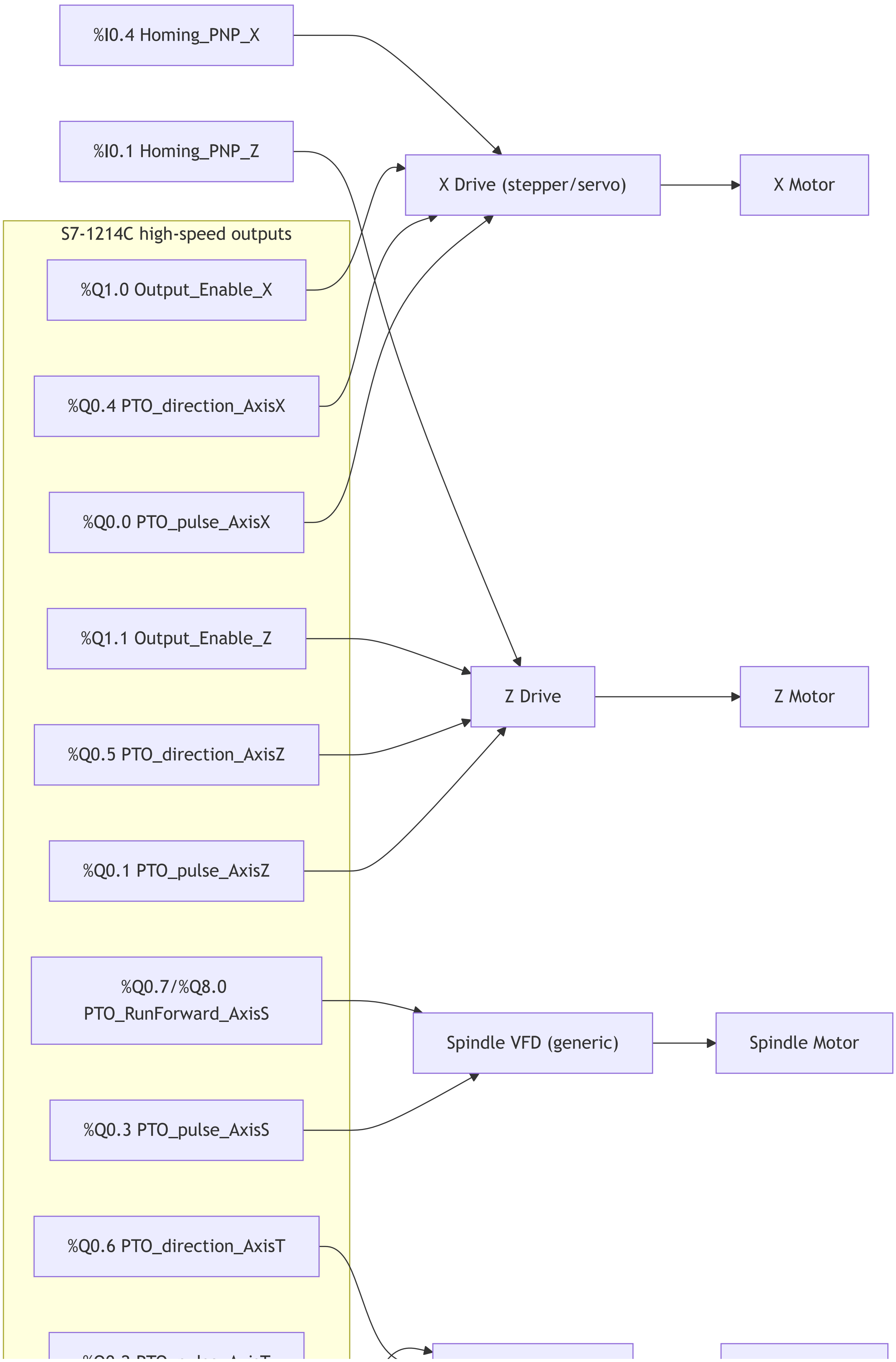
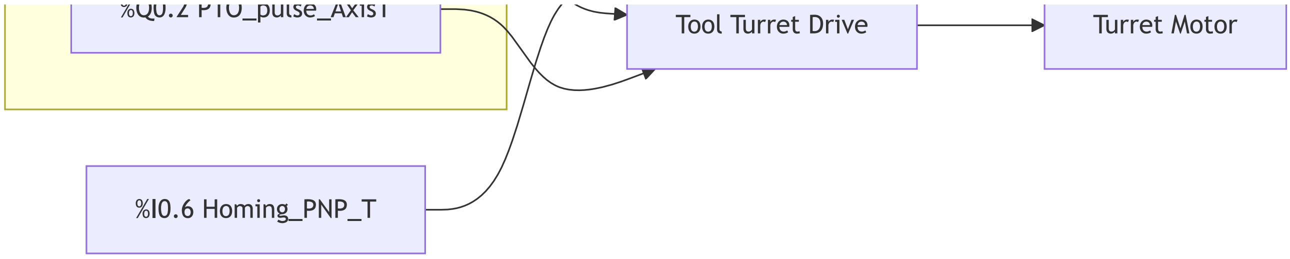
Sheet 2 — Motion Drives (X / Z / Tool / Spindle)

⚠ Drives and VFD are generic. Match PTO output type (24 V/5 V, sink/source) to the drive's pulse/direction inputs; add level shifters/opto isolation as the drive requires.


%I0.4 Homing_PNP_X
%I0.1 Homing_PNP_Z
S7-1214C high-speed outputs
%Q1.0 Output_Enable_X
%Q0.4 PTO_direction_AxisX
%Q0.0 PTO_pulse_AxisX
%Q1.1 Output_Enable_Z
%Q0.5 PTO_direction_AxisZ
%Q0.1 PTO_pulse_AxisZ
%Q0.7/%Q8.0 PTO_RunForward_AxisS
%Q0.3 PTO_pulse_AxisS
%Q0.6 PTO_direction_AxisT
X Drive (stepper/servo)
X Motor
Z Drive
Z Motor
Spindle VFD (generic)
Spindle Motor



Notes

- Axes X, Z, Tool are PTO positioning axes (pulse + direction). Each homes to a PNP reference prox.
- **Spindle is velocity mode:** PLC sends a pulse train (`%Q0.3`) as the frequency reference plus a level **RunForward** enable. Direction is CW only (no CCW output on this machine). ⚠️ Do **not** wire the spindle so that losing the pulse train faults the VFD on a normal stop — by design the PLC keeps pulsing and uses RunForward to start/stop (see project note `project_spindle_pto_keep_pulsing`). Configure the VFD accordingly.
- `Output_Enable_X/Z` are separate drive-enable lines; Tool/Spindle have no separate enable output.
- ⚠️ `PTO_RunForward_AxisS` appears in the tag export at **both** `%Q0.7` and `%Q8.0` (`PTO_RunForward_AxisS(1)`). Confirm the single correct terminal during commissioning and remove the duplicate tag.
- Encoder/feedback wiring (if servo) goes drive↔motor and, where used, drive↔PLC per the drive manual — not represented here.

Signal	Address	Device
PTO_pulse_AxisX	%Q0.0	X drive pulse
PTO_direction_AxisX	%Q0.4	X drive direction
PTO_pulse_AxisZ	%Q0.1	Z drive pulse
PTO_direction_AxisZ	%Q0.5	Z drive direction
PTO_pulse_AxisT	%Q0.2	Tool drive pulse
PTO_direction_AxisT	%Q0.6	Tool drive direction
PTO_pulse_AxisS	%Q0.3	Spindle VFD pulse/frequency reference
PTO_RunForward_AxisS	%Q0.7 / %Q8.0	Spindle VFD run-forward enable (confirm one)
Output_Enable_X	%Q1.0	X drive enable
Output_Enable_Z	%Q1.1	Z drive enable
Homing_PNP_X	%I0.4	X homing reference proximity (PNP NO)
Homing_PNP_Z	%I0.1	Z homing reference proximity (PNP NO)
Homing_PNP_T	%I0.6	Tool homing reference proximity (PNP NO)

Sheet 3 — PLC Digital I/O Terminal Map

Digital Inputs (%I)

Address	Tag	Device / Contact	Logic
%I0.0	Limit_PNP_Z_Min	Z min soft-limit prox	PNP NO — TRUE in zone
%I0.1	Homing_PNP_Z	Z homing reference prox	PNP NO
%I0.2	Limit_PNP_Z_Max	Z max soft-limit prox	PNP NO — TRUE in zone
%I0.3	Limit_PNP_X_Min	X min soft-limit prox	PNP NO — TRUE in zone
%I0.4	Homing_PNP_X	X homing reference prox	PNP NO
%I0.5	Limit_PNP_X_Max	X max soft-limit prox	PNP NO — TRUE in zone
%I0.6	Homing_PNP_T	Tool homing reference prox	PNP NO
%I0.7	Limit_NC_Z_Min	Z min hard limit switch	**NC** — opens at limit
%I1.0	Limit_NC_Z_Max	Z max hard limit switch	**NC** — opens at limit
%I1.1	Limit_NC_X_Min	X min hard limit switch	**NC** — opens at limit
%I1.2	Limit_NC_X_Max	X max hard limit switch	**NC** — opens at limit
%I1.4	Safety_Estop_Ch1	E-Stop channel 1	**NC** — closed = safe
%I1.5	Safety_Estop_Ch2	E-Stop channel 2	**NO** — open = safe
%I8.0	Panel_Start_A	Start button A (two-hand)	NO momentary
%I8.1	Panel_Start_B	Start button B (two-hand)	NO momentary
%I8.2	In_Cyl_ToolHeadLock_AtSetpoint	Tool-head-lock magnetic sensor	NO — TRUE = locked
TBD	Safety_Door	Door closed switch	NC recommended
TBD	Safety_Air	Air pressure switch	NO — TRUE = OK
TBD	Panel_Stop / Pause / Reset	Panel buttons (or HMI)	NO momentary

Digital Outputs (%Q)

Address	Tag	Device	Notes
%Q0.0	PTO_pulse_AxisX	X drive pulse	high-speed
%Q0.1	PTO_pulse_AxisZ	Z drive pulse	high-speed
%Q0.2	PTO_pulse_AxisT	Tool drive pulse	high-speed
%Q0.3	PTO_pulse_AxisS	Spindle VFD pulse	high-speed
%Q0.4	PTO_direction_AxisX	X direction	
%Q0.5	PTO_direction_AxisZ	Z direction	
%Q0.6	PTO_direction_AxisT	Tool direction	
%Q0.7	PTO_RunForward_AxisS	Spindle run-forward	duplicate of %Q8.0 — confirm one
%Q1.0	Output_Enable_X	X drive enable	
%Q1.1	Output_Enable_Z	Z drive enable	
%Q8.0	PTO_RunForward_AxisS(1)	Spindle run-forward (dup)	confirm vs %Q0.7
%Q8.4	Output_Contactor_Spindle	Spindle contactor coil	
%Q8.5	Output_Contactor_X	X contactor coil	
%Q8.6	Output_Contactor_Z	Z contactor coil	
%Q8.7	Output_Contactor_Tool	Tool contactor coil	
%Q12.0	Output_Cyl_Backsupport_SolA	BackSupport extend solenoid	5/3 BC
%Q12.1	Output_Cyl_Backsupport_SolB	BackSupport retract solenoid	5/3 BC
%Q12.2	Output_Cyl_SheetHolder_SolA	SheetHolder extend solenoid	5/2 SR
%Q12.3	Output_Cyl_SheetHolder_SolB	SheetHolder (2nd sol, see note)	tag exists; DB ValveType=1
%Q12.4	Output_Cyl_ToolHeadLock_SolA	ToolHeadLock lock solenoid	5/2 SR (safety)
%Q12.5	Output_Cyl_MandrelLock_SolA	MandrelLock clamp solenoid	5/2 SR
%Q12.7	Output_Cyl_Backsupport_SolAtmosphere	BackSupport vent solenoid	CMD=41 pressure relief

Same map as a Graphviz DOT diagram

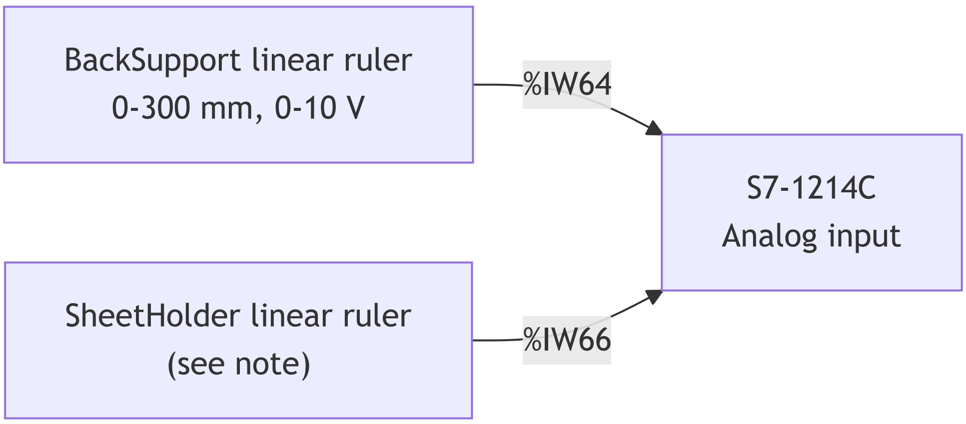
```
digraph PLC_IO {
    rankdir=LR;
    node [shape=box, fontname="Helvetica"];

    subgraph cluster_in { label="Field Inputs"; color=gray;
        E_Stop; Limits_NC; Prox_PNP; Homing; Start_AB; LockSensor; Door_Air [label="Door / Air (TBD)"];
    }
    PLC [shape=box3d, label="S7-1214C"];
    subgraph cluster_out { label="Field Outputs"; color=gray;
        Drives [label="Drives X/Z/Tool (PTO+dir+en)"];
        VFD [label="Spindle VFD (pulse+RunFwd)"];
        Contactors [label="Contactors %Q8.4-8.7"];
        Valves [label="Solenoid valves %Q12.x"];
    }

    E_Stop      -> PLC [label="%I1.4 NC / %I1.5 NO"];
    Limits_NC   -> PLC [label="%I0.7,%I1.0-1.2 NC"];
    Prox_PNP    -> PLC [label="%I0.0,0.2,0.3,0.5"];
    Homing      -> PLC [label="%I0.1,0.4,0.6"];
    Start_AB    -> PLC [label="%I8.0,%I8.1"];
    LockSensor  -> PLC [label="%I8.2"];
    Door_Air    -> PLC [label="TBD"];

    PLC -> Drives     [label="%Q0.0-0.2,0.4-0.6,1.0-1.1"];
    PLC -> VFD        [label="%Q0.3,%Q0.7/8.0"];
    PLC -> Contactors [label="%Q8.4-8.7"];
    PLC -> Valves     [label="%Q12.0-12.5,12.7"];
}
```

Sheet 4 — Analog I/O (Linear Rulers)

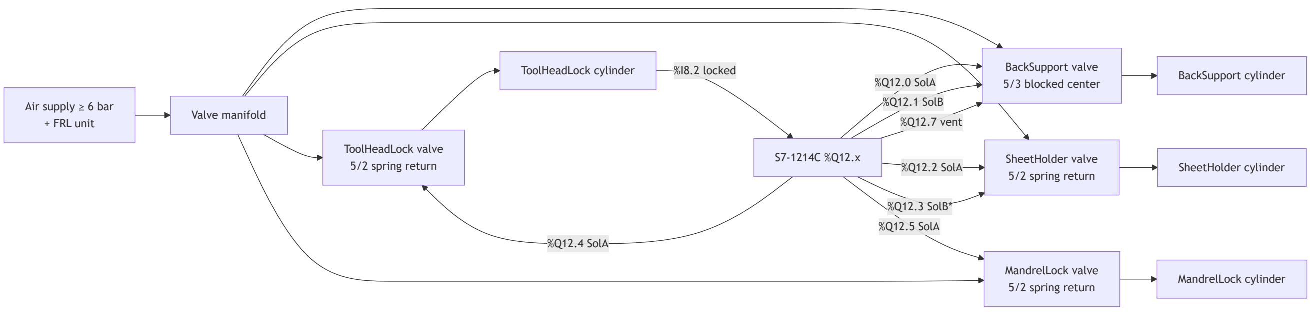


Signal	Address	Device	Scaling
AI_LinearRuler_Backsupport	%IW64	BackSupport position sensor (0–10 V)	Raw 0–31433 → 0–300 mm, **signal inverted** (0 V = extended ~300 mm, ~10 V = retracted 0 mm). Calibrate `Raw_Max` after install.
AI_LinearRuler_SheetHolder	%IW66	SheetHolder position sensor	Tag present in export; ruler feedback currently not used by the SheetHolder DB (PositioningMode=0). Wire only if feedback will be enabled.

Note: The BackSupport runs in timed full-stroke mode (PositioningMode=0); the ruler input is read but ignored unless PositioningMode is changed to 3. Wire the ruler so calibration can be performed.

Sheet 5 — Pneumatic Solenoid Manifold

⚠️ All cylinder valve types and behaviors are taken from `Program/02_DataBlocks.scl` and `Program/09_Sensors_Actuators.scl`. Confirm valve hardware matches before connecting.



Cylinder	Valve	Solenoid(s)	Fail-safe (power loss)	Position feedback
BackSupport	5/3 blocked center (ValveType=2)	SolA %Q12.0 (extend), SolB %Q12.1 (retract), Atmosphere %Q12.7 (vent)	Holds last position (blocked center)	Analog ruler %IW64 (timed mode, feedback unused)
SheetHolder	5/2 spring return (ValveType=1)	SolA %Q12.2 (extend/hold)	Spring retracts = safe	None (timed full stroke). %Q12.3 SolB tag exists but DB is single-solenoid — leave unwired unless re-configured.*
ToolHeadLock	5/2 spring return (ValveType=1)	SolA %Q12.4 (lock)	Spring retracts = **unlocked (safe)**	Magnetic sensor %I8.2 (confirms locked); 6 s timeout → error
MandrelLock	5/2 spring return (ValveType=1)	SolA %Q12.5 (clamp)	Spring retracts	None (timed full stroke)

Safety behavior notes

- **ToolHeadLock** is safety-critical: energized (locked) only in RUNNING/PAUSED; spring-unlocks in all other states and on power loss. The magnetic sensor (%I8.2) must confirm the lock within 6 s or the machine faults (0x0501 path / cylinder error).
- **MandrelLock** is held clamped during spindle coast-down (RUNNING/PAUSED/STOPPING/ERROR) even if E-Stop fires, to prevent the spinning blank from ejecting; it releases via an explicit retract

pulse on Ack/Reset.
- **BackSupport** blocked-center valve keeps the cylinder in place when both solenoids are off; the atmosphere/vent solenoid (%Q12.7) is used for pressure relief via recipe command CMD=41.

* The SheetHolder `SolB` (%Q12.3) and a SheetHolder ruler (%IW66) appear in the tag export but the current DB configures SheetHolder as a single-solenoid, timed (no-sensor) cylinder. Treat these as reserved/future and confirm with Maintenance before wiring.

Commissioning checklist

- [ ] Resolve all **TBD** addresses (Safety_Door, Safety_Air, Panel_Stop/Pause/Reset).
- [ ] Resolve the duplicate **PTO_RunForward_AxisS** (%Q0.7 vs %Q8.0) to one terminal.
- [ ] Verify NC limit and E-Stop contacts open the circuit on fault (fail-safe), not close it.
- [ ] Confirm PTO output electrical level matches each drive's pulse/direction input.
- [ ] Configure the spindle VFD so a held pulse train + RunForward = run; normal stop must not fault the VFD (PTO keeps pulsing by design).
- [ ] Calibrate the BackSupport ruler `Raw_Max` (`02_DataBlocks.scl` / FC_LoadConfig Section 6).
- [ ] Verify the ToolHeadLock magnetic sensor (%I8.2) triggers within 6 s of lock command.
- [ ] Verify contactor drop-out on E-Stop is hardware-independent of the PLC output.

Source of truth: `Program/docs/PLCTags.xlsx` (addresses) and `Program/08_Main_OB1.scl`, `Program/09_Sensors_Actuators.scl`, `Program/02_DataBlocks.scl` (signal usage and valve config). Generic power/drive blocks must be confirmed against the installed panel.